



Meta-analysis of the effects of sulfate versus hydroxy trace mineral source on nutrient digestibility in dairy and beef cattle

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ABSTRACT

Trace mineral (TM) source can potentially alter nutrient digestibility through effects on microbial populations. A meta-analysis was conducted to determine whether sulfate versus hydroxy (IntelliBond) sources of supplemental Cu, Zn, and Mn had any effect on dry matter intake (DMI), dry matter digestibility, and neutral detergent fiber (NDF) digestibility. All available cattle studies (8 studies, 12 comparisons) were used to estimate the effect size (hydroxy mean – sulfate mean). Factors included in the analysis were method of digestibility analysis (total collection, marker-based, or 24 h in situ), study design (randomized design or Latin square), beef ($n = 5$) versus dairy ($n = 7$) cattle, and days on treatment; these factors were retained when $P < 0.05$. Dry matter digestibility was increased by hydroxy TM in beef (1.64 ± 0.35 units) but not in dairy models (0.16 ± 0.13 units) relative to sulfate TM. The NDF digestibility increased significantly with hydroxy versus sulfate TM, but digestibility assessment method influenced this response. Studies using total collection or undigested NDF as a flow marker showed a significant increase (2.68 ± 0.40 units and 1.08 ± 0.31 units, respectively) in NDF digestibility for hydroxy versus sulfate TM; but studies utilizing 24-h in situ incubation did not detect any change (-0.03 ± 0.23 units). These observations may reveal differences in precision of measurement or may indicate mineral effects beyond the rumen; total collection is considered the gold standard method. Hydroxy TM did not affect DMI per animal or per unit of body weight relative to sulfate TM. In conclusion, feeding hydroxy versus sulfate TM does not appear to affect DMI but, depending on type of cattle and method of measurement, can increase dry matter digestibility and NDF digestibility, which may be explained by differences in solubility of the TM sources in rumen, differentially affecting fermentation.

Key words: mineral, ruminant, fiber

INTRODUCTION

Minerals are inorganic elements which, similar to other nutrients, are essential for physiological functions and need to be replenished mainly through the diet to avoid health issues. Major functional roles of minerals are to provide structure for the body, contribute to regulatory mechanisms, and catalyze various enzymatic reactions in the body (Suttle, 2010). Macrominerals are required in greater quantities (e.g., grams per day), whereas trace minerals (TM) are required in significantly lesser quantities (e.g., milligrams per day). Typically, only 0.01% of an organism's total mass is constituted by TM (Rabiee et al., 2010). Nevertheless, TM are required and essential for biochemical processes supporting growth, maintenance, milk production, fertility, and immunity (Hilal et al., 2016).

Adequate TM concentrations in feed are often taken for granted because TM effects on animal physiology are routinely underestimated (López-Alonso, 2012). This demonstrates the importance of considering TM concentrations in diets, solubilities, and expected bioavailabilities. Bioavailability, in turn, can be affected by a myriad of factors including source of TM, TM chemical characteristics and solubility (Miller et al., 2020), dietary characteristics such as dietary NDF (Kabaija and Smith, 1988), and levels of antagonists in the diet and gut environment (Overton and Yasui, 2014). The inverse is also true; differences in solubility of TM can modulate digestibility of other nutrients by affecting ruminal microflora (Caldera et al., 2019). Common forms of TM used to supplement cattle include sulfates, oxides, carbonates, chlorides, hydroxy, and organic complexes (Daniel et al., 2020). In the past, the livestock industry has relied mostly on oxide TM sources before shifting to sulfate TM sources, although more recently there is some evidence pointing to benefits of TM supplementation using more bioavailable inorganic or organic TM sources (Overton and Yasui, 2014). Copper, Zn, and Mn are the most common TMs supplemented for dairy cows, deficiencies of which may lead to various health problems. Copper is required for functioning of numerous enzymes and cofactors and its

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deficiency in cattle can lead to salt-sickness, severe diarrhea, and bone disorders such as widening of epiphyses (Suttle, 2010). Similarly, Zn deficiency can result in severe parakeratosis, anorexia, inappetence, bone disorders, and infertility in cattle (Miller and Miller, 1962; Pitts et al., 1966; Suttle, 2010). Manganese deficiency can reduce conception rates and cause delayed estrous and fetal deformities (Hansen et al., 2006).

Given the potential effects of TM such as Cu, Zn, and Mn, solubility of different TM sources at various gastrointestinal sites may be an important consideration. According to Caldera et al. (2019), solubility of Cu and Zn hydroxy TM at a ruminal pH of 6.23 was lesser than that of sulfates of these minerals. Also, addition of Zn sulfate and Cu sulfate to a washed suspension of ruminal microorganisms reduced cellulose digestion, although Mn sulfate did not have an effect (Hubbert et al., 1958). In vitro, as little as 1.5 mg/L Cu and 5 mg/L Zn (wt:vol in rumen fluid) inhibited cellulose digestion (Hubbert et al., 1958). Less soluble TM sources in the rumen may result in decreased bioavailability of Zn and Cu to cellulolytic microorganisms (Daniel et al., 2020), potentially leading to lesser interference. Hence, the hypothesis of our meta-analysis is that hydroxide-based TM sources result in greater NDF digestibility (**NDFD**), DM digestibility, and DMI when compared with sulfates of Cu, Mn, and Zn. The objective of the meta-analysis is to determine whether there is evidence in the literature that TM sources of Cu, Zn, and Mn affect NDF and DM digestibility or DMI.

MATERIALS AND METHODS

Literature Search

Studies and reports, including published and unpublished reports, were considered for inclusion in the meta-analysis. Inclusion of unpublished studies in the meta-analysis may help to counter possible publication bias because there is evidence that research studies with nonsignificant results are less likely to be published (Sterling, 1959). Publication bias is a common problem that can significantly alter results of effects being investigated (Thornton and Lee, 2000). An attempt was made to ensure inclusion of all studies evaluating sulfate versus hydroxy TM sources in the meta-analysis by conducting a literature search between July and September 2021. A single investigator carried out the search, seeking reports published from 1990 through the time of the search; older studies were excluded in part due to different NDF analysis methodology before 1990 and partly due to expected differences in passage rate in less productive cattle with lesser DMI.

Google Scholar (<https://scholar.google.com/>) was searched using the following search terms: trace mineral and source and sulfate and digestibility and (hydroxychloride or IntelliBond). Similarly, Scopus (<https://www.scopus.com/>) was searched using the terms: trace mineral and source and digestibility and (cow or cattle). We also sought reports from Micronutrients USA (Indianapolis, IN); at the time of this analysis there were no other data sets (to our knowledge) in the public domain using hydroxy TM produced by other companies. Of the 7 reports provided by Micronutrients, 5 were published in peer-reviewed journals and 2 were unpublished reports. After exclusion of duplicate results, 78 reports were obtained.

Studies were then screened for inclusion and were retained only if cattle were fed identical basal diets with sulfate and hydroxy TM added in equal quantities (mineral basis). Only studies containing at least some measure of digestibility while comparing the hydroxy and sulfate TM sources in cattle were retained. One full text was excluded after recognition that it reported ruminal digestibility from a study with total-tract digestibility data already included in the database. After screening (Figure 1), 8 unique reports were found to meet criteria for inclusion in the meta-analysis. The 8 reports provided a total of 12 comparisons, shown in Table 1.

Data Extraction

The factors included to account for possible effects on results included method of digestibility analysis, study design, beef versus dairy cattle use, TM feeding rates, days on TM treatment, intake as percentage of BW, and percentage NDF in diet. Also included was the standard error of the mean (**SEM**) for response variables and the mean difference in the response variables (hydroxy TM mean – sulfate TM mean). For standardization, the 24-h time point was used for studies where in situ methods were used for digestibility assessment. One (Daniel et al., 2020) of the 4 studies categorized using the undigested NDF (**uNDF**) marker for digestibility assessment used a 336-h incubation to define uNDF, whereas the remaining 3 studies used a 240-h incubation. Although this difference may have led to trivial differences in absolute digestibility results, the relative differences between treatments within study were likely unaffected, and all uNDF marker studies were considered to have used a similar method.

When additional data on diets or other study factors was required, authors were contacted to request the data ($n = 1$). Only digestibilities of DM and NDF were available from all the studies, whereas ad libitum DMI

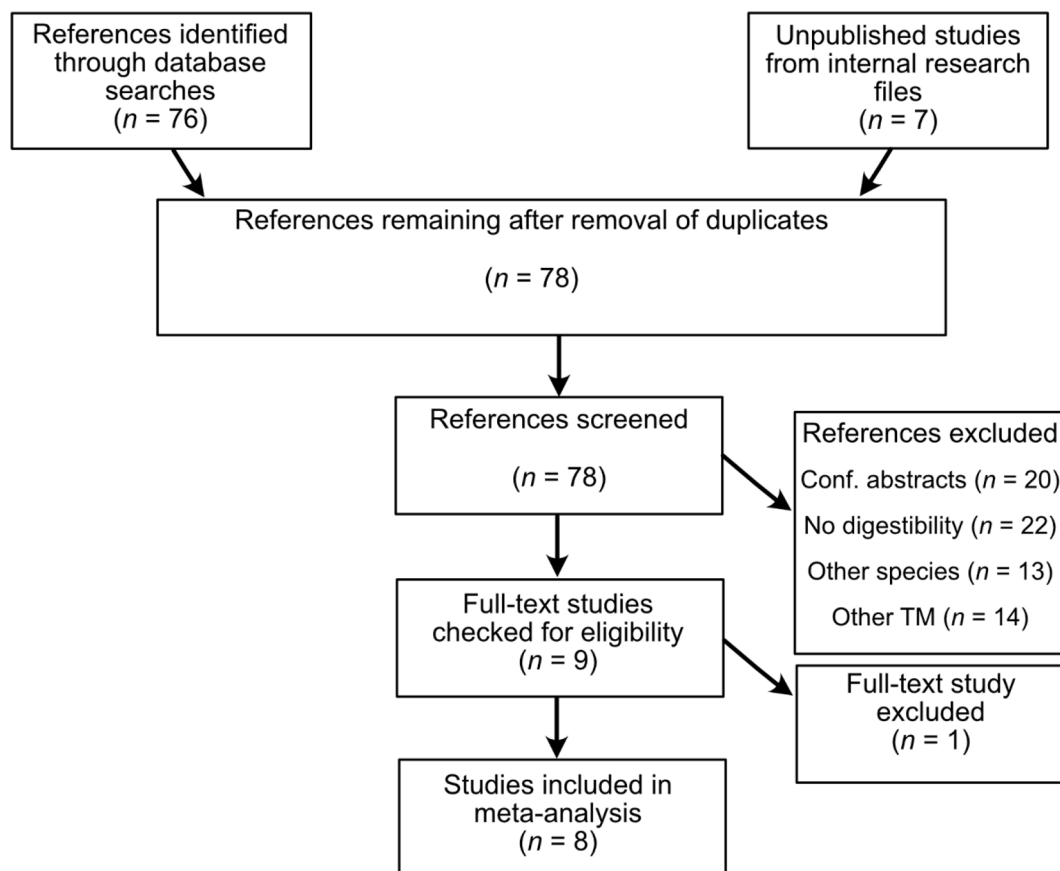


Figure 1. Preferred Reporting Items for Systemic Reviews and Meta-Analysis (PRISMA) flow diagram showing the stages from initial search to finalization of publications included in the meta-analysis evaluating effects of trace mineral (TM) source supplementation on DM digestibility, NDF digestibility, and DMI.

Table 1. List of studies, comparisons, and explanatory variables included in the meta-analysis

Reference	Study type	Design code	Observations per treatment	Method ¹	Top dress or TMR	Beef or dairy
Caldera et al., 2019	Randomized block	RCBD	4	Total collection	Top dress	Beef
Daniel et al., 2020	Latin square	LS	13	uNDF ₃₃₆	TMR	Dairy
Faulkner and Weiss, 2017 (byproduct)	Split-plot replicated Latin square	LS	18	Total collection	Top dress	Dairy
Faulkner and Weiss, 2017 (forage)	Split-plot replicated Latin square	LS	18	Total collection	Top dress	Dairy
Genther and Hansen, 2015 (low level)	5 × 5 Latin square	LS	5	In situ 24 h	TMR	Beef
Genther and Hansen, 2015 (high level)	5 × 5 Latin square	LS	5	In situ 24 h	TMR	Beef
Guimaraes et al., 2021	Randomized block	RCBD	6	Total collection	TMR	Beef
Guimaraes et al., 2022	Randomized block	RCBD	4	Total collection	TMR	Beef
Miller et al., 2020 (conventional)	Replicated 4 × 4 Latin square	LS	14	uNDF ₂₄₀	TMR	Dairy
Miller et al., 2020 (brown mid-rib)	Replicated 4 × 4 Latin square	LS	14	uNDF ₂₄₀	TMR	Dairy
Micronutrients trial 2017D123USCZM	Crossover	LS	14	uNDF ₂₄₀	TMR	Dairy
Micronutrients trial 2017D103CACZM	Split-plot replicated Latin square	LS	6	In situ 24 h	TMR	Dairy

¹uNDF = undigested NDF after incubation in rumen fluid for either 336 or 240 h.

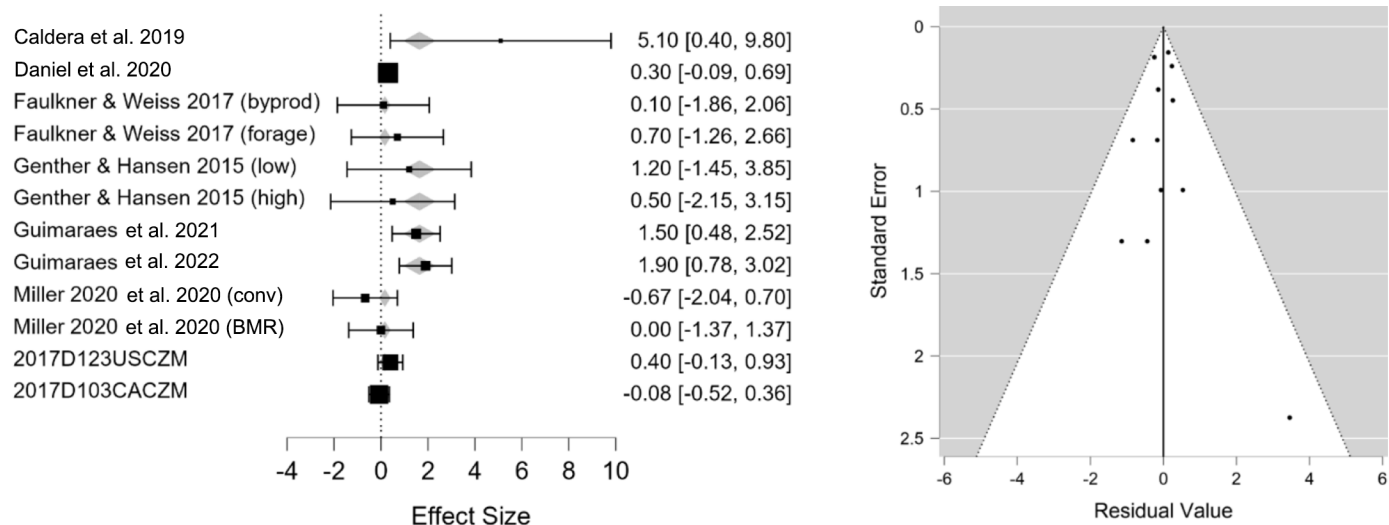


Figure 2. Forest plot and funnel plot for DM digestibility (%) response by animal type for hydroxy trace mineral (TM) versus sulfate TM sources. Individual study means are represented by squares in the forest plot, with the subcategory (beef and dairy) means shown by diamonds; error bars show the 95% confidence intervals.

response was available for 9 of the 12 comparisons. Studies and comparisons are listed in Table 1.

Statistical Analysis

Data were analyzed with a mixed-model approach using SAS software (v. 9.4; SAS Institute Inc.), treating study as a random effect and other explanatory variables tested as fixed effects in assessing the mean effect size for hydroxy versus sulfate TM. All results are defined as the response to hydroxy TM (hydroxy mean – sulfate mean). Studies were weighted based on the inverse of the squared SEM for the outcome of interest. Funnel plots (JASP v. 0.12.2; Figures 2, 3, 4, and 5) were used to evaluate the potential for publication bias in the data set, and the Test of Residual Heterogeneity (Q test) was used to assess whether substantial between-study variation in response existed. When the Q test was significant, predictive variables were added to the model in a forward stepwise process and retained when $P < 0.05$. When predictive variables were retained in the model, biologically relevant interactions were tested; in these cases, both main effects were retained in the model. No interactions were retained in final models.

RESULTS

Dry Matter Digestibility

Dry matter digestibility data were available from 12 treatment comparisons derived from a total of 268

individual animal observations. We found no evidence for an overall effect on DMD (+0.50 units, SEM = 0.27, $P = 0.11$; Table 2) when hydroxy TM replaced sulfate TM. However, assessment of the model revealed significant heterogeneity ($Q = 22.49$, $P = 0.02$) across studies. After evaluating multiple explanatory factors, sector (beef vs. dairy) was retained as the only significant factor in the model ($P = 0.01$), and after accounting for sector, residual heterogeneity was no longer significant ($Q = 7.36$, $P = 0.69$). The results showed a significant mean increase of 1.64 units (SEM = 0.35, $P < 0.001$; Table 3) in DMD when hydroxy versus sulfate TM were consumed by beef animals. However, for dairy cattle a nonsignificant mean increase of 0.16 units (SEM = 0.13, $P = 0.29$; Table 3) in DMD was observed. The forest and funnel plots for the final DM digestibility model are shown in Figure 2. The funnel plot did not show any evidence of observation bias (Egger's test of asymmetry: $P = 0.96$), suggesting that a representative sampling of outcomes is included in the analysis.

Neutral Detergent Fiber Digestibility

The NDFD data from the 12 treatment comparisons derived from a total of 242 individual animal observations were included in the statistical model. Overall, NDFD increased significantly (1.51 units, SEM = 0.49, $P = 0.02$; Table 2) in response to hydroxy versus sulfate TM. As in the case of DMD, however, model assessment again showed heterogeneity ($Q = 33.09$, $P < 0.001$) across studies. After testing all study factors in the model, digestibility assessment method was retained as

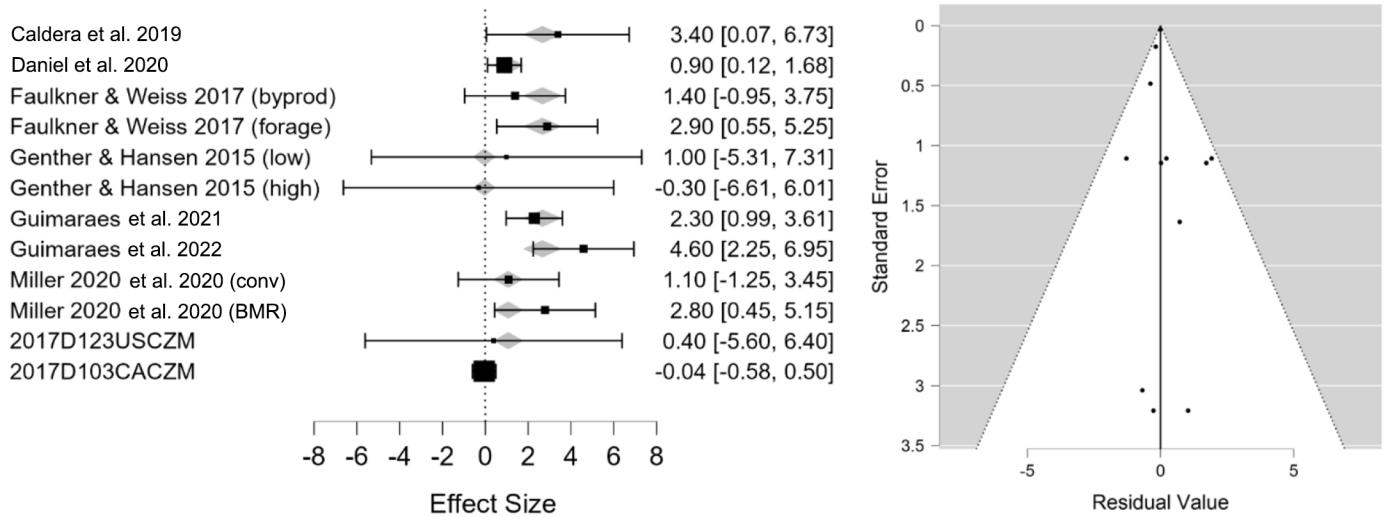


Figure 3. Forest plot and funnel plot for NDF digestibility (%) response by digestibility method for hydroxy trace mineral (TM) versus sulfate TM sources. Individual study means are represented by squares in the forest plot, with the subcategory (digestibility method) means shown by diamonds; error bars show the 95% confidence intervals.

the only significant factor in the final model ($P = 0.01$), and residual heterogeneity was not significant anymore ($Q = 6.65$, $P = 0.67$). Studies using the total collection method and uNDF as a marker showed a significant mean increase of 2.68 units ($SEM = 0.40$, $P < 0.01$; Table 4) and 1.08 units ($SEM = 0.31$, $P = 0.02$) in NDFD for hydroxy versus sulfate TM, respectively. However, studies using the 24-h in situ incubation method did not detect a significant change in NDFD (-0.03 units, $SEM = 0.23$; $P = 0.89$) across TM sources.

We considered possible confounding of digestibility method and animal type in the evaluation of factors affecting NDFD effect size, despite the failure of sector (beef vs. dairy) to be retained in the NDFD model. The greatest mean NDFD response was observed for total collection methods ($P < 0.04$ vs. other methods), which were used in 2 dairy and 3 beef animal studies, and in situ methods were used in 1 dairy and 2 beef studies. Use of uNDF as a digestibility marker was the only method that was not represented in both sectors, as

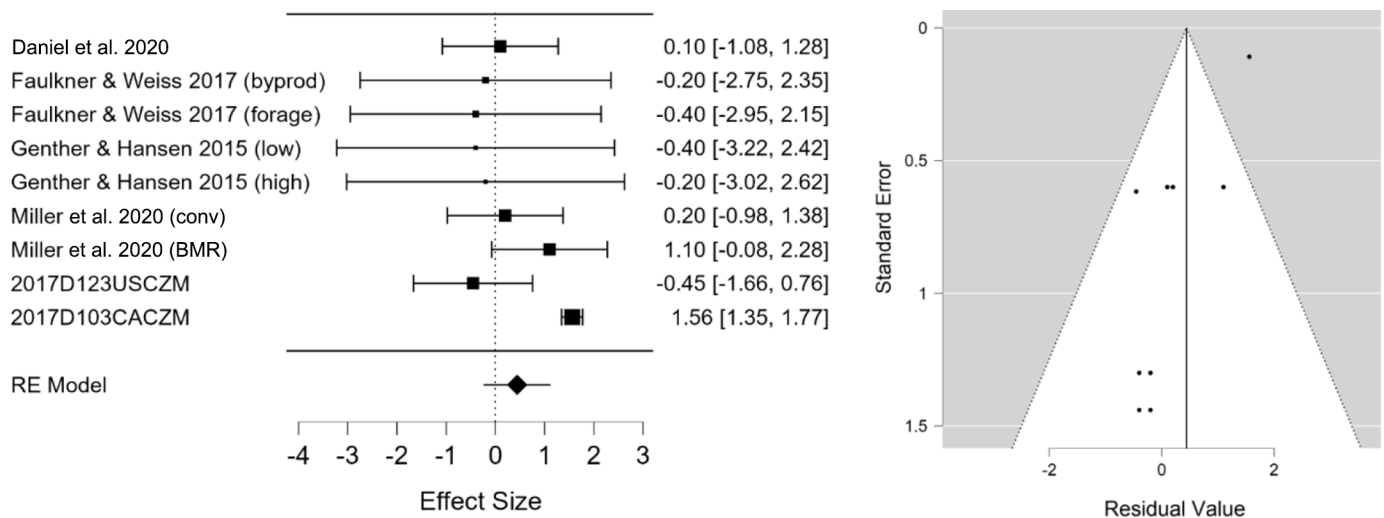


Figure 4. Forest plot and funnel plot for DMI (kg/d) response for hydroxy trace mineral (TM) versus sulfate TM sources. Individual study means are represented by squares with the overall mean shown by the diamond; error bars show the 95% confidence intervals. RE model = overall effect size estimate from the restricted maximum likelihood (mixed effects) model.

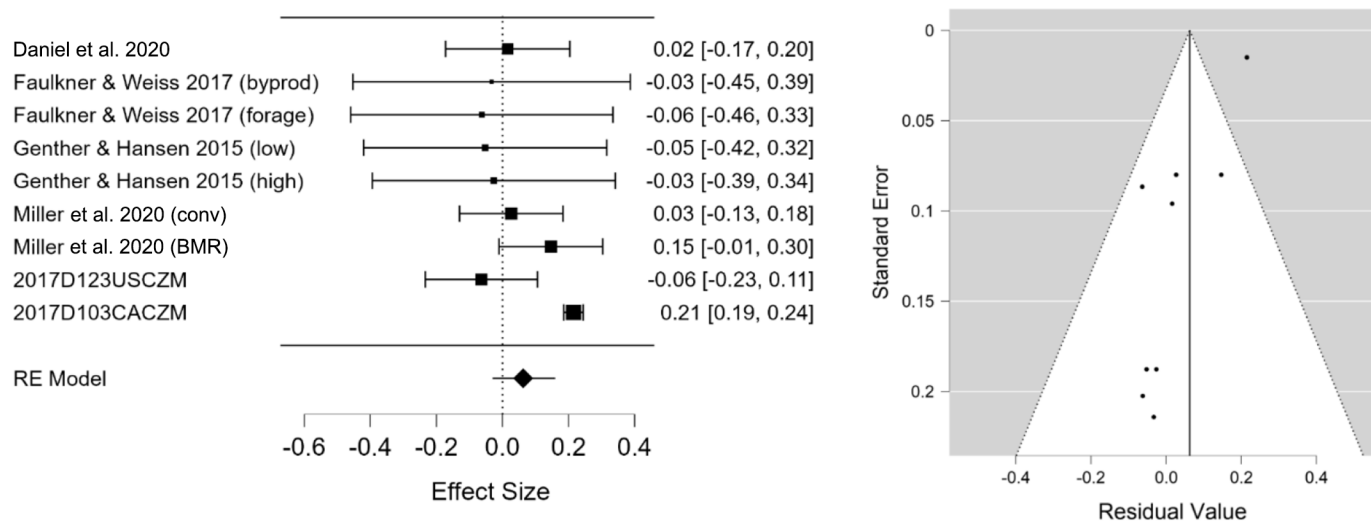


Figure 5. Forest plot and funnel plot for DMI (% of BW) response for IntelliBond versus sulfate mineral sources. Individual study means are represented by squares with the overall mean shown by the diamond; error bars show the 95% confidence intervals. RE model = overall effect size estimate from the restricted maximum likelihood (mixed effects) model.

it was used in dairy cattle studies ($n = 4$) and no beef studies. We found no evidence of digestibility method $\times$ sector interaction in the model for NDFD response ($P = 0.72$), suggesting that differences in outcomes were not due to confounding. Forest and funnel plots for the final NDFD model are shown in Figure 3. The funnel plot again revealed no evidence of observation bias (Egger's test of asymmetry: $P = 0.41$).

Dry Matter Intake

For DMI, 9 out of 12 available comparisons were used in the meta-analysis, as DMI was restricted to 90% of ad libitum intake in 3 of the studies. Overall, DMI per animal (kg/d) and DMI per unit bodyweight (%BW) showed a nonsignificant mean response of +0.30 kg/d (SEM = 0.35, $P = 0.43$; Table 2) and +0.04% BW (SEM = 0.048, $P = 0.47$), respectively, for hydroxy versus sulfate TM. Heterogeneity was observed for this variable when expressed both in kg/d ($Q = 26.29$, $P < 0.001$) and as a % of BW ($Q = 208.64$, $P < 0.001$), as shown in Table 2. However, no additional factors were retained in the model, so the response heterogeneity

was not resolved with the data available. The distribution of responses for DMI metrics are represented in forest and funnel plots (Figure 4 and 5). Unlike the other outcomes evaluated, the Egger's test of asymmetry ($P < 0.001$ for both funnel plots) indicated an unexpected effect size distribution. The removal of study 2017D103CACZM resulted in funnel plots with more normal distributions (not shown). However, the conclusions were not altered by the inclusion or exclusion of this outlier study, as the mean effect size for DMI did not differ from 0 in either case (all $P > 0.40$).

DISCUSSION

Hydroxychloride-based TM in comparison to sulfate TM significantly increased DMD in beef cattle but not in dairy cattle studies. The difference in DMD response across sectors was not explained by variation in DMI (% of BW) or NDF content of the diets. So, there is little evidence to suggest that an interaction of TM source with either passage rate or dietary fiber content explains the observed different responses between beef and dairy cattle models, although the limited number

Table 2. Simple least squares means estimates of responses to replacing sulfate trace minerals (TM) with hydroxy TM and test of response heterogeneity

Outcome	Comparisons (n)	Mean response	SEM	P-value (treatment means)	Q	P-value (Q)
DM digestibility (%)	12	+0.50	0.27	0.11	22.5	0.02
NDF digestibility (%)	12	+1.51	0.49	0.02	33.1	<0.001
DMI (kg/d)	9	+0.30	0.35	0.43	26.3	<0.001
DMI (% of BW)	9	+0.04	0.049	0.44	24.7	<0.01

Table 3. Dry matter digestibility (%) response by animal type for hydroxy trace minerals (TM) versus sulfate TM sources

Sector	Comparisons (n)	Mean response	SEM	P-value
Beef	5	+1.64	0.35	<0.001
Dairy	7	+0.16	0.13	0.29

of treatment means available for this analysis limits conclusions regarding this lack of explanatory power. Perhaps the easiest explanation for a more dramatic effect of TM source on DMD in beef cattle is the greater total TM concentrations in the beef cattle diets in these studies. Among the 3 beef and 6 dairy cattle comparisons that provided total dietary concentrations of the treatment minerals (including basal diet supply), Cu, Zn, and Mn concentrations were on average 64, 55, and 25% greater, respectively, in the beef cattle studies. Although supplemental Mn amounts did not differ between beef and dairy studies, supplemental Cu and Zn supplies (mg/d) were 31 and 18% greater, respectively, in the beef cattle studies (all 12 comparisons). Combined with the expected lesser mass of digesta in the rumen of beef versus dairy cattle, the steady-state concentration of these minerals was likely greater in beef cattle studies included, which could help to explain why the beef cattle studies found an effect of TM source on DMD that was not detected in dairy cattle.

Although there was a slight overall effect of TM source on DMD, an overall significant increase of 1.51 units in NDFD was evident for hydroxy versus sulfate TM supplementation. Heterogeneity across studies for NDFD response to TM source revealed digestibility assessment method as the only significant factor in the model. Variation in NDFD response to TM source observed across studies using the 3 different methods seemed to align with method precision. Total fecal collection is the gold standard procedure (Lee and Hristov, 2013) for accurate nutrient digestibility assessment, thus the significant effect of TM source in studies utilizing this method strengthens confidence in the overall finding of improved NDFD for hydroxy versus sulfate TM.

Total collection methods can be impractical and labor intensive, and indigestible markers are often used as an alternative approach for estimating fecal output. Significant interactions between intrinsic markers and diets are not uncommon, although uNDF has been more reliable than acid-insoluble ash for dairy digestibility studies (Lee and Hristov, 2013; Daniel et al., 2020). In our analysis, uNDF methodology was confounded with animal type, as all 4 studies using this method were dairy animal studies, which may have contributed to the observed mean. However, method by animal type interaction was not evident in the model for NDFD ($P = 0.72$; other methods included both animal types),

Table 4. Neutral detergent fiber digestibility (%) response by digestibility assessment method for hydroxy trace minerals (TM) versus sulfate TM sources

Method	Comparisons (n)	Mean response	SEM	P-value
24-h in situ	3	-0.03	0.23	0.89
Total collection	5	+2.68	0.40	<0.01
uNDF marker ¹	4	+1.08	0.31	0.02

¹uNDF= undigested NDF after incubation in rumen fluid for either 336 h (uNDF₃₃₆) or 240 h (uNDF₂₄₀). One of the studies used uNDF₃₃₆, whereas the other 3 used uNDF₂₄₀ as the marker method.

and animal type was not retained in the model. Similar results between total collection and uNDF-based methods here suggest reasonable agreement in terms of effects of TM source on total-tract NDFD.

In situ incubation methods failed to yield similar results, which can perhaps be attributed to the models being noisy, making it harder to detect relatively small differences in NDFD in response to TM source. Factors affecting 24-h in situ results include in situ bag location within the rumen, rinsing technique used (Weiss, 1994), pore size, and basal diet effect on microbial populations migrating inside the bag (Meyer and Mackie, 1986). Beyond lesser precision, the 24-h in situ method could potentially also miss effects of TM source on postruminal digestion. According to Firkins (1997), typical rations having NDF content of 25 to 30% with a higher ratio of potentially digestible NDF will result in increased hindgut fiber fermentation. In dairy cattle, fermentation in the large intestine has been reported to account for anywhere from 4 to 23% of total-tract NDF digestion (Gressley et al., 2011), and TM effects in the hindgut cannot be ruled out, given the relatively low total bioavailability of TM (NASEM, 2021). Therefore, there is a real possibility that 24-h in situ methods do not accurately capture the effect of TM source on NDFD, both ruminally and postruminally.

Neither metric of DMI was significantly affected by TM source (Table 2). According to Genther and Hansen (2014), the TM profile of beef steers was a critical factor in maintaining DMI during disease or shipping stress. In a separate beef steer TM supplementation level study by Niedermayer et al. (2018), no mean difference in DMI (kg/d) on d 0 to 56 was reported, however inorganic TM supplementation from d 56 to 124 increased DMI compared with control, though not among varying levels of TM supplementation. This perhaps indicates that long-term studies of TM sources might yield more conclusive results for DMI. Usually a 14-d adaptation period is considered adequate for changeover and crossover experiments (Machado et al., 2016), although a range of 7 to 27 d adaptations are not unheard of for digestibility trials. The average length of treatment in the 9 comparisons used for DMI

evaluation was 21 d, but 4 of these comparisons used 12-d periods.

Extent of TM solubility determines the concentration of metallic minerals exposed to rumen microbes, which may be an important factor determining its interaction with microbes (Genther and Hansen, 2015). For example, Cu from hydroxy-based TM was found to be less soluble in the rumen but had similar solubility as sulfates in the acidic abomasum (Genther and Hansen, 2015). Copper is known to be toxic to cellulolytic bacteria even at 1.5 mg/L in solution (Hubbert et al., 1958). The selective solubility depending on gut compartment environment is likely one way that hydroxy-based TM differ from sulfates. It is also worth considering the sulfate component itself. In a diet containing Cu, Zn, and Mn sulfates to provide 15 mg/kg Cu, 50 mg/kg Zn, and 25 mg/kg Mn, the TM supplements would provide approximately 51 mg/kg sulfur that would not be found in an equivalent hydroxy TM blend. However, in comparison to the dietary sulfur recommendation of 0.2% of DM (2,000 mg/kg; NASEM, 2021), this additional sulfur amounts to just a 2.6% increment in supply, making this an unlikely factor driving differential responses between TM sources.

CONCLUSIONS

Hydroxy TM compared with sulfate TM positively affected DMD in beef but not in dairy animals. Hydroxy versus sulfate TM also showed an overall increase in NDFD; however, digestibility assessment method affected response, with the largest effect observed for total collection studies followed by those using uNDF as a marker, whereas those using 24-h in situ methods did not detect any NDFD response. Because total collection methods are widely considered the gold standard for measuring digestibility, the results give credence to the TM source effect. Overall, digestibility responses to different TM sources may dependent on cattle type, digestibility assessment methods, and properties of the mineral source itself, but these findings point to potential for improved digestibility with hydroxy versus sulfate TM.

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